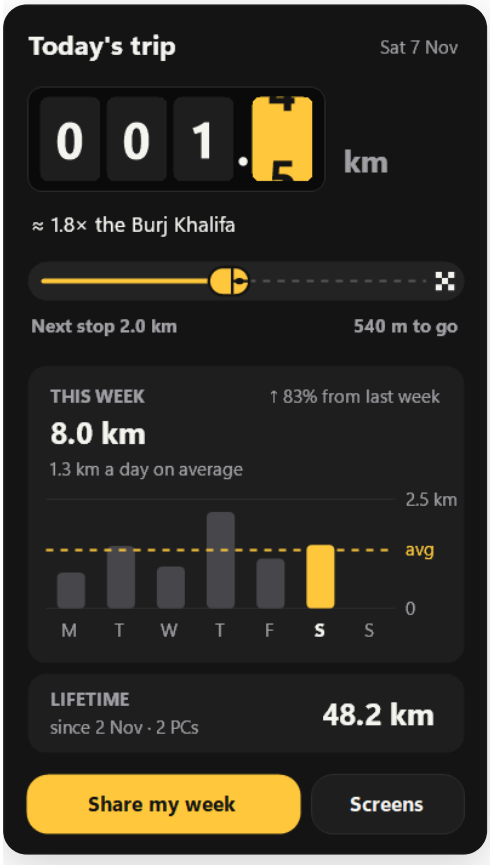
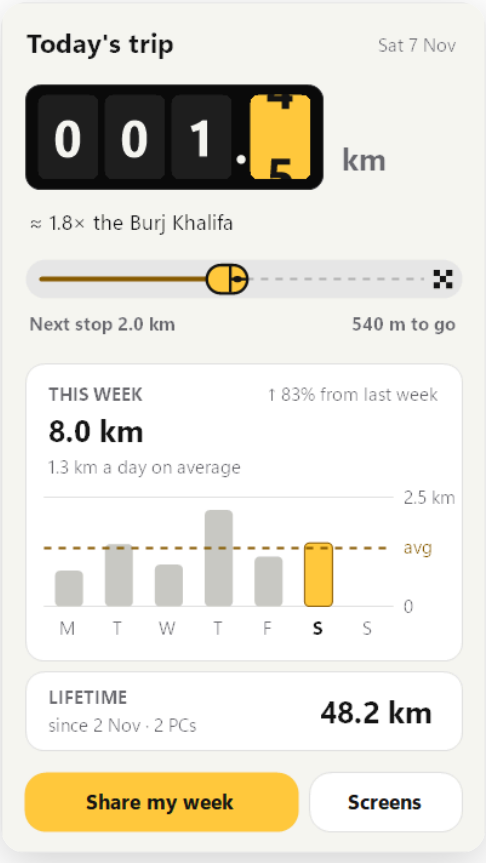
How Mouse Travel works

A tiny Windows tray app that turns your mouse movement into meters and kilometers, shown on a rolling car odometer. This guide walks through everything it does, from the moment Windows starts to the numbers you see.





The big picture

Every part of the app follows one loop: start up, learn your screens, measure the cursor, save safely, and show you the result.

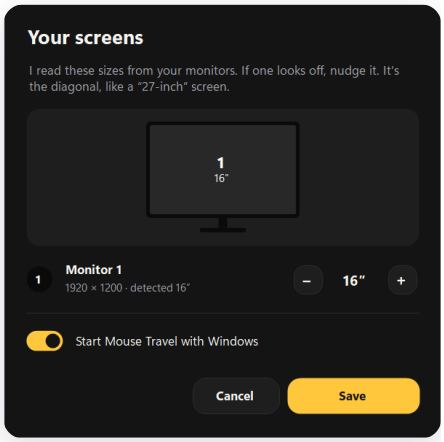
Start — Opens at sign-in from the Startup folder → **Screens** — Learns each monitor's real size → **Measure** — Cursor position 25× a second → **Save** — Every minute, in several safe places → **Show** — Popup, pop-ups, weekly card

1 Starting up

- When you sign in, Windows opens the **"Mouse Travel" shortcut in your Startup folder**, which runs `MouseTravel.exe`.
- Only **one copy** can run at a time. Launching it again just quits quietly.
- It loads your saved data, works out today's date, puts the yellow lane-driver icon in the tray, and saves once right away (which also refreshes your history file).

2 Knowing your screens

- Each monitor reports its physical size to Windows (from its built-in EDID info). Mouse Travel reads it: your laptop is **16"** and the Dell is **27"**.
- From size and resolution it calculates **how many millimetres one pixel is** on each screen. A pixel on the 27″ Dell is bigger than one on the laptop.
- If a monitor doesn't report its size, the **Screens** window lets you set it with − / +. Plugging in or removing a monitor makes it check again automatically.



The Screens window draws your monitors to scale.

3 Measuring distance

- **25 times a second**, it asks Windows where the cursor is. That's all it reads: **no clicks, no keystrokes**.
- If the cursor moved, it measures the straight-line distance in pixels, converts it using that screen's mm-per-pixel, and adds it to **today's total**.
- It measures **how far the cursor travels on screen**, at real-world size. It's deliberately a rough, fun number, not your hand's exact movement.

4 Saving your data

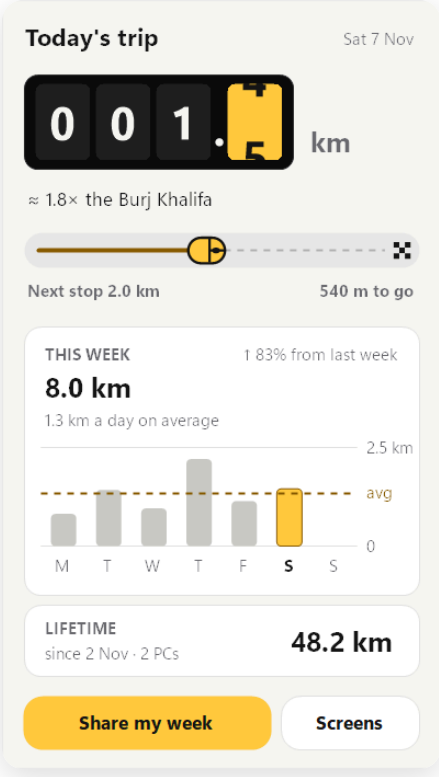
Once a minute, plus at startup and when Windows shuts down, Mouse Travel saves to several places:

Where	What
`%APPDATA%\MouseTravel\days.csv`	The main data: one line per day, exact meters
`Mouse Travel history.txt` (next to the exe)	Your readable notepad copy: every day, week and lifetime. Also read back as a backup
`%APPDATA%\MouseTravel\backup\`	One copy per day, keeping the last 7 days
`My Drive\Mouse Travel\<PC name>.csv`	Only when **Sync across PCs** is turned on

> **Safety rule:** before every save, the app re-reads all of these and keeps the **highest** number for each day. A day's distance can only go up, so a bad startup can never wipe your history. A small `log.txt` records every load and save.

5 What you see

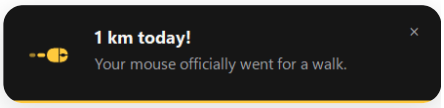
- **Tray icon:** the lane-driver logo. Hover over it for Today, This week and Lifetime. It can show this week's km instead (right-click menu).
- **Left-click** opens the popup:
  - **Odometer:** today's km on rolling wheels. They roll up from where you last looked and keep creeping while you move.
  - **The road:** the mouse drives toward your next milestone, with "540 m to go".
  - **This week**, Screen Time style: total, daily average, "↑ x% from last week", and Mon–Sun bars with a dashed avg line (today in yellow).
  - **Lifetime** since your first day ("· 2 PCs" when synced), plus **Share my week** and **Screens**.
- **Right-click** menu: share this week or last week, Screens..., Start with Windows, Sync across PCs..., show this week's km on the icon, open data folder, Quit.
- It follows Windows **light/dark mode** and the **animation** setting.



The popup in light mode (sample numbers).

6 Milestone pop-ups

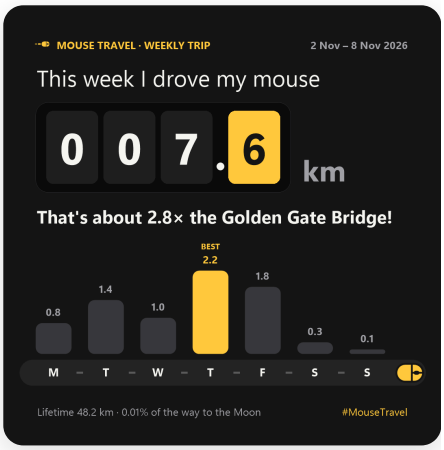
- **Today:** 500 m, 1 km, then every km. **Lifetime:** 5 km, 10 km, half marathon, marathon, then 100 km and up.
- They're Mouse Travel's **own card** in the bottom-right corner, so they work even with Windows notifications turned off.
- They **never steal focus**, pause while you hover over them, close after 5 seconds or on click, and **wait** while you're in a full-screen app.
- Each milestone fires once (once per day for daily ones), even across restarts.



A milestone pop-up. The yellow line counts down 5 seconds.

7 Weekly share card

- Every Monday a pop-up says “Last week: X km”. Click it, or use **Share my week**, to get a 1080×1080 image.
- The card shows the odometer, a comparison (“2.8× the Golden Gate Bridge”), daily bars with your **best** day, and your lifetime.
- It's saved to `Pictures\Mouse Travel` and copied to your clipboard, ready to paste into WhatsApp or Instagram.



The weekly share card (sample week).

8 Sync across PCs (optional)

- Right-click the tray icon, choose **Sync across PCs...**, and pick a Google Drive or OneDrive folder (it suggests `G:\My Drive`). Do the same on each PC.
- Each PC writes only **its own file** there and reads the others'. That means **no sync conflicts**, and Today, Week and Lifetime **add up all your computers**.
- A reset PC gets its history back as soon as you point it at the same folder. **Stop syncing** appears in the same menu.

9 Under the hood

- **One C# file** (`src/MouseTravel.cs`), built with the compiler that **ships with Windows**, so there's nothing to install. The exe is about **76 KB** and uses a few MB of memory.
- **Privacy:** everything stays on your PC (and your own Google Drive if you turn on sync). Nothing is sent anywhere.
- **Updates:** new versions are published on GitHub as releases with the exe attached.